Structurally Sparse Molecular Hamiltonians for Near-Term Quantum Simulation

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We present a library of qubit Hamiltonians for molecular quantum simulation that are structurally sparser than conventional Gaussian-basis alternatives by one to three orders of magnitude. The Hamiltonians are constructed from a spectral graph theory framework (GeoVac) that exploits natural coordinate separations in molecular electronic structure, producing block-diagonal electron repulsion integrals and Gaunt-enforced selection rules that persist through Jordan-Wigner encoding. For the composed architecture (PK-partitioned), LiH requires 334 Pauli terms at 30 qubits versus 907 for Gaussian STO-3G at 12 qubits, with 13× fewer qubitwise-commuting measurement groups. The balanced coupled architecture (PK-free) achieves 0.20% energy error at the equilibrium geometry of LiH when solved exactly, with 878 Pauli terms at 30 qubits. Across six first-row molecules (LiH, BeH_2, H_2O, HF, NH_3, CH_4), within-molecule Pauli scaling follows $O(Q^{2.2})$, compared to $O(Q^{3.9-4.3})$ for Gaussian baselines. The Hamiltonians are available as a Python library with Qiskit, OpenFermion, and PennyLane export.

I. INTRODUCTION

Quantum simulation of molecular electronic structure is among the most promising near-term applications of quantum computing [1–3]. However, the cost of implementing molecular Hamiltonians on quantum hardware is dominated by the number of Pauli terms in the Jordan-Wigner (or equivalent) encoding: each distinct Pauli string requires separate measurement circuits, and the total number of shots scales with the square of the Pauli 1-norm [4, 5].

For a molecule encoded in Q qubits using a Gaussian atomic orbital basis and the Jordan-Wigner transformation, the Pauli term count scales as $O(Q^4)$ due to the dense four-index electron repulsion integrals (ERIs). Post-hoc compression techniques—double factorization [6], tensor hypercontraction [7], qubit tapering [8]—can reduce this cost but do not alter the underlying structure of the Hamiltonian.

An alternative is to construct Hamiltonians that are *structurally* sparse: the ERIs are block-diagonal by construction, with zeros enforced by angular momentum selection rules rather than discovered by compression. The GeoVac framework achieves this by exploiting the natural coordinate separations of molecular electronic structure [9–12]. Each electron group (core, bond pair, lone pair) is described in its own natural coordinate system, coupled via cross-center nuclear attraction integrals and cross-block Coulomb integrals. The angular matrix elements are computed from Gaunt integrals (Wigner $3j$ symbols), which impose exact selection rules that translate directly into structural zeros in the qubit Hamiltonian.

The result is Pauli scaling of $O(Q^{2.2-2.5})$ across first-row molecules, with 51–1,712× fewer terms than Gaussian raw Jordan-Wigner baselines at comparable qubit counts [10]. This structural sparsity is compatible with all downstream optimizations (tapering, grouping, tensor factorization) and is basis-intrinsic rather than compression-dependent.

This paper presents benchmark data for the GeoVac Hamiltonian library: accuracy characterization via full configuration interaction (FCI), quantum resource comparison against Gaussian baselines, and installation instructions. Theory derivations are deferred to the supporting papers [9–12].

II. THE HAMILTONIAN

A. What the user receives

A GeoVac Hamiltonian is a standard qubit operator—a weighted sum of Pauli strings—delivered as a `SparsePauliOp` (Qiskit), `QubitOperator` (OpenFermion), or `qml.Hamiltonian` (PennyLane):

```
from geovac.ecosystem_export import
    hamiltonian
h = hamiltonian('LiH', R=3.015, max_n=2)
op = h.to_qiskit()    # 334 Pauli terms
op_of = h.to_openfermion()
op_pl = h.to_pennylane()
print(h.n_qubits)     # 30
print(h.one_norm)      # 33.3 Ha
```

The operator is ready for use in any VQE, ADAPT-VQE, QPE, or other quantum algorithm that consumes a Pauli Hamiltonian.

B. Two architectures

GeoVac provides two Hamiltonian architectures for each molecule:

a. Composed (PK-partitioned). The Phillips-Kleinman (PK) pseudopotential, a one-body operator that enforces core-valence orthogonality, is separated from the qubit Hamiltonian and evaluated classically from the VQE one-particle reduced density matrix

(1-RDM) [11]:

$$E_{\text{total}} = \langle\psi|H_{\text{elec}}|\psi\rangle + \text{Tr}(h^{\text{PK}}\,\gamma)\,. \tag{1}$$

This produces the sparsest qubit operator (334 Pauli terms for LiH) at the cost of requiring a classical post-processing step.

b. Balanced coupled (PK-free). Cross-center nuclear attraction integrals, computed via a multipole expansion that terminates exactly at $L = 2\ell_{\max}$ by Gaunt selection rules [12], replace the PK pseudopotential. This produces a larger operator (878 Pauli terms for LiH at $n_{\max} = 2$) but one that is variationally complete within the composed partition—FCI on this Hamiltonian gives physical energies without any classical correction.

C. How it works (brief)

Electrons are partitioned into groups (core, bond pairs, lone pairs) based on the chemical structure. Each group is described in its natural coordinate system (hyperspherical for core electrons, molecule-frame hyperspherical for bond pairs). Within each group, the angular basis states are labeled by hydrogenic quantum numbers (n, ℓ, m), and all angular matrix elements are evaluated from Gaunt integrals—products of spherical harmonics integrated analytically via Wigner $3j$ symbols. The selection rules $|\Delta\ell| \leq k$, $|\Delta m| \leq k$ produce exact structural zeros in the ERI tensor that persist through Jordan-Wigner encoding. Full derivations are in Refs. [9–12].

III. ACCURACY BENCHMARKS

A. LiH potential energy surface

We characterize the balanced coupled Hamiltonian by computing 4-electron FCI energies across the LiH potential energy surface. Table I shows the FCI energy at seven bond lengths for $n_{\max} = 2$ ($Q = 30$ qubits, 15 spatial orbitals, FCI dimension 11,025).

TABLE I. LiH balanced coupled FCI energies at $n_{\max} = 2$ ($Q = 30$, 878 Pauli terms, analytical V_{ne} integrals). E_{exact} is the non-relativistic Born-Oppenheimer energy [13].

R (bohr)	E_{FCI} (Ha)	ΔE (mHa)	Error (%)
2.0	−7.8246	181	2.3
2.5	−7.8940	164	2.1
3.0	−7.9290	142	1.8
3.015	−7.9294	141	1.8
3.5	−7.9273	137	1.7
4.0	−7.8952	157	2.0
5.0	−7.7716	254	3.3

The FCI-predicted equilibrium is $R_{\text{eq}} = 3.227$ bohr (7.0% error vs. experimental 3.015 bohr). At $n_{\max} = 3$ ($Q = 84$, 42 spatial orbitals, FCI dimension 741,321), the energy at $R = 3.015$ bohr improves to −8.0545 Ha (0.20% error), and the equilibrium shifts to $R_{\text{eq}} = 3.28$ bohr (8.8% error). Table II summarizes the basis convergence.

TABLE II. Basis convergence: LiH balanced coupled at $R = 3.015$ bohr. Exact non-relativistic energy: −8.0706 Ha.

$n_{\max}$	Q	E_{FCI} (Ha)	Error (%)	R_{eq} (bohr)	R_{eq} err
2	30	−7.929	1.8	3.227	7.0%
3	84	−8.055	0.20	3.28	8.8%

The 9× improvement in energy error from $n_{\max} = 2$ to $n_{\max} = 3$ demonstrates rapid basis convergence. The equilibrium geometry drifts by +0.053 bohr per unit increase in $n_{\max}$, approximately 3× smaller than the drift observed with the PK pseudopotential [12]. The energy error at $n_{\max} = 3$ (0.20%) is competitive with Gaussian double-zeta bases for single-point calculations.

B. Multi-geometry FCI: solver vs. Hamiltonian accuracy

The 5.3–26% equilibrium geometry errors reported in the classical solver literature for GeoVac [11] arise from the classical solver (adiabatic approximation, PK overcounting), not from the Hamiltonian operator itself. When the balanced coupled Hamiltonian is solved exactly via FCI:

- The energy error at equilibrium drops from 15% (composed + PK) to 1.8% ($n_{\max} = 2$) or 0.20% ($n_{\max} = 3$).
- The balanced coupled configuration is the *only* one that produces a bound LiH molecule in 4-electron FCI—decoupled (no cross terms), composed with PK, and coupled without cross-center V_{ne} all give unbound results [12].
- The variational bound is respected at all geometries tested ($E_{\text{FCI}} > E_{\text{exact}}$ everywhere).

C. BeH_2 and H_2O

For BeH_2 ($Q = 50$), the balanced coupled 4-electron FCI gives $E = -14.146$ Ha (10.7% error), compared to −11.497 Ha (27.4%) for composed with PK. The balanced 1-norm (304.7 Ha) is 14% lower than composed (354.9 Ha), demonstrating that the PK-free architecture is cheaper for quantum simulation.

For H_2O ($Q = 70$), FCI is infeasible (10 electrons), but the Hamiltonian metrics are available: 5,798 Pauli terms, 1,509 Ha 1-norm, 168 QWC groups. The balanced 1-norm is 0.054× the composed total (which is dominated by the PK barrier at $Z_{\text{eff}} = 6$), and 4.2× the composed electronic-only 1-norm.

D. Applicability: single-point vs. dissociation

The balanced coupled Hamiltonian is designed for single-point energy calculations at fixed molecular geometries—the use case for quantum resource estimation, VQE benchmarks, and QPE demonstrations. For this primary use case, the balanced coupled Hamiltonian achieves 0.20% energy accuracy at $n_{\max} = 3$.

The Hamiltonian is *not* designed for computing dissociation curves or reaction energies between different molecular species. The cross-center V_{ne} multipole expansion includes a monopole ($1/R$) component that decays slowly, causing the electronic energy to remain unconverged even at $R = 100$ bohr. The practical dissociation energy computed from the PES well depth ($D_e = 0.161$ Ha at $n_{\max} = 2$) overestimates the experimental value (0.092 Ha) by 74%. Users requiring dissociation energies should use Gaussian-basis methods for the dissociation limit and GeoVac Hamiltonians for single-point calculations at experimentally known or DFT-optimized geometries.

IV. QUANTUM RESOURCE COMPARISON

A. Pauli terms and measurement cost

Table III compares GeoVac Hamiltonians against Gaussian raw Jordan-Wigner baselines.

The comparison is most favorable for LiH, where the GeoVac composed Hamiltonian has 2.7× fewer Pauli terms and 13× fewer QWC measurement groups than Gaussian STO-3G, with a nearly identical electronic 1-norm (33.3 vs. 34.3 Ha). Against larger Gaussian bases, the advantage grows: 190× fewer Pauli terms than cc-pVDZ.

For H_2O, the composed Hamiltonian has *more* Pauli terms than STO-3G (778 vs. 551, advantage 0.71×) because the GeoVac encoding uses 70 qubits (15 spatial orbitals per block × 5 blocks × 2 spins) while STO-3G uses only 12 qubits (7 spatial orbitals with frozen core). The meaningful comparison is at comparable basis quality: against cc-pVDZ (46 qubits), GeoVac wins by 138×. The $O(Q^{2.2})$ vs. $O(Q^4)$ scaling guarantees that the GeoVac advantage increases with basis size.

B. Scaling exponents

Table IV shows the within-molecule Pauli scaling exponents, obtained from two-point fits at $n_{\max} = 1$ and $n_{\max} = 2$.

The GeoVac composed scaling exponent of ~ 2.2 is nearly constant across all six molecules, reflecting the universal Gaunt selection rule structure. The Gaussian scaling exponent of ~ 4 reflects the dense four-index ERI tensor. This difference means the GeoVac advantage grows polynomially with system size: at $Q = 100$, the expected Pauli ratio is $(100)^{4.1}/(100)^{2.2} \approx 6{,}000\times$.

C. Regime dependence

The GeoVac advantage is strongest in the NISQ/VQE regime where Pauli term count and QWC measurement groups dominate the runtime cost. For fault-tolerant QPE, the relevant cost metric is the Pauli 1-norm $\lambda = \sum_j |c_j|$, which determines the Trotter step count. At matched accuracy (GeoVac balanced $n_{\max} = 3$ vs. Gaussian cc-pVTZ), the 1-norm comparison is less favorable due to the larger qubit count of the GeoVac encoding (84 vs. 54 qubits for LiH). However, published double-factorization or THC λ values for molecules at $Q < 100$ do not exist in the literature [6, 7], making a direct 1-norm comparison at the QPE operating point infeasible.

V. AVAILABLE SYSTEMS

Table V lists all molecules currently available in the GeoVac Hamiltonian library.

A. Installation and usage

The library is available at `https://github.com/jloutey-hash/geovac` and can be installed via:

```
pip install geovac-hamiltonians
```

or from source:

```
git clone https://github.com/jloutey-hash/
    geovac
pip install -e .
```

Ecosystem compatibility: Qiskit ≥1.0, OpenFermion ≥1.6, PennyLane ≥0.35. Python ≥3.10.

VI. LIMITATIONS

a. Composed partition. The Hamiltonian is constructed within a composed partition that assigns each electron to a specific group (core, bond, lone pair). This is an approximation of the full N-electron Hilbert space—inter-group correlations beyond those captured by the cross-block Coulomb and cross-center nuclear attraction integrals are neglected. For LiH, the full 4-electron treatment in the same angular basis gives 63.5% R_{eq} error (angular basis limited), while the composed partition gives 7.0–8.8% (basis more efficient but inter-group antisymmetry approximate) [11].

TABLE III. Quantum resource comparison: GeoVac vs. Gaussian raw JW. GeoVac composed values use PK classical partitioning (electronic-only 1-norm). Gaussian data: computed from OpenFermion (STO-3G) and Trenev et al. [14] (6-31G, cc-pVDZ). "—" indicates data not available.

System	Method/Basis	Q	Pauli terms	1-norm (Ha)	QWC groups	Advantage[a]
5*LiH	GeoVac composed	30	334	33.3	21	—
	GeoVac balanced	30	878	74.1	87	—
	Gaussian STO-3G	12	907	34.3	273	2.7× Pauli, 13× QWC
	Gaussian 6-31G	20	5,851	—	—	17.5× Pauli
	Gaussian cc-pVDZ	36	63,519	—	—	190× Pauli
4*H_2O	GeoVac composed	70	778	361	21	—
	GeoVac balanced	70	5,798	1,509	168	—
	Gaussian STO-3G	12	551	—	—	0.71× Pauli
	Gaussian cc-pVDZ	46	107,382	—	—	138× Pauli
2*BeH_2	GeoVac composed	50	556	67.4	21	—
	GeoVac balanced	50	2,652	304.7	138	—

[a]Ratio of Gaussian to GeoVac composed Pauli terms. Values > 1 favor GeoVac; values < 1 favor Gaussian.

TABLE IV. Within-molecule Pauli scaling: $N_{\text{Pauli}} \propto Q^{\alpha}$. GeoVac exponents from composed architecture (2-point fit, $n_{\max} = 1, 2$). Gaussian exponent from STO-3G/6-31G/cc-pVDZ progression.

System	GeoVac α	Gaussian α
LiH	2.24	3.9–4.3
BeH_2	2.18	—
H_2O	2.22	3.9–4.3
HF	2.21	—
NH_3	2.20	—
CH_4	2.19	—
Mean	2.21 ± 0.02	3.9–4.3

TABLE V. GeoVac composed Hamiltonian library. All values at $n_{\max} = 2$ with PK classical partitioning (electronic-only metrics shown).

Molecule	Q	Pauli terms	1-norm (Ha)	QWC groups
LiH	30	334	33.3	21
BeH_2	50	556	67.4	21
H_2O	70	778	361	21
HF	60	667	—	—
NH_3	80	889	—	—
CH_4	90	1,000	—	—

b. Basis completeness. At $n_{\max} = 2$ (the production operating point for composed Hamiltonians), single-point energies have 1.8% error for LiH. At $n_{\max} = 3$, this improves to 0.20%, but at the cost of 84 qubits and 19,959 Pauli terms (balanced coupled) or 7,879 (composed). The $n_{\max} = 3$ basis approaches chemical accuracy for single-point energies but not for potential energy surfaces (8.8% R_{eq} error persists due to structural drift in the composed partition).

c. Dissociation limit. The dissociation limit is not converged (see Sec. III D).

d. Classical solver accuracy. The classical (non-quantum) solver results reported in Refs. [11, 12] (5.3–26% R_{eq} error) reflect classical solver limitations, not Hamiltonian quality. FCI on the balanced coupled Hamiltonian gives substantially better energies (0.20% at $n_{\max} = 3$), validating the Hamiltonian for quantum simulation use.

e. Scope. First-row atoms ($Z = 1$–10) are fully supported. Second-row atoms are feasible with external frozen-core data but not yet implemented. Transition metals are out of scope due to the angular basis truncation at $\ell_{\max} = 2$ [15].

f. Qubit count. GeoVac Hamiltonians use more qubits than Gaussian alternatives at the same basis quality (30 vs. 12 for LiH at STO-3G–equivalent accuracy). The advantage is in Pauli count and measurement cost, not qubit economy. For near-term hardware where measurement budget (not qubit count) is the bottleneck, this tradeoff is favorable.

VII. CONCLUSION

The GeoVac Hamiltonian library provides structurally sparse qubit Hamiltonians for first-row molecular quantum simulation. The structural sparsity—$O(Q^{2.2})$ Pauli scaling from Gaunt selection rules—is an intrinsic property of the basis, not an artifact of compression. For LiH, this yields 2.7× fewer Pauli terms and 13× fewer measurement groups than Gaussian STO-3G, growing to 190× fewer terms against cc-pVDZ. The balanced coupled architecture achieves 0.20% energy accuracy at FCI, demonstrating that the Hamiltonian is suitable for high-accuracy quantum simulation at fixed geometries.

Future directions include: (i) quantum resource esti-

mation for fault-tolerant algorithms using GeoVac Hamiltonians, (ii) hardware demonstrations on trapped-ion or superconducting platforms, (iii) compatibility benchmarks with tensor factorization and active-space reduction techniques, and (iv) extension to second-row molecules.

The library is open-source and pip-installable. All Hamiltonians presented in this paper are reproducible from the public repository.

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